

Wealth-Voyager: Navigating IntelligentWealth Management with a Multi-Agent Framework

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Abstract

Rising demand for hyper-personalized wealth management is increasingly unmet by traditional advisory models, which suffer from limitations in scalability, cost-efficiency, and objectivity. While recent AI-driven solutions show promise, they often remain fragmented, focusing on isolated tasks and lacking a unified architecture that integrates long-term strategy with dynamic behavioral adaptation. To bridge this gap, we introduce **Wealth-Voyager**, a multi-agent framework that synergizes strategic asset allocation with adaptive tactical adjustments. Our system's architecture uniquely combines **AlphaForge**, a quantitative engine for establishing a long-term strategic portfolio, with **DualAdvisor**, a novel Belief-Desire-Intention (BDI) grounded role-playing simulation that interactively manages user-specific behavioral biases in response to real-time events. Our proof-of-concept case study, conducted under live market conditions, demonstrated the efficacy of this integrated approach. The system's adaptive tactical adjustments outperformed a passive baseline, converting a marginal gain into a more substantial return while reducing portfolio volatility. Our work presents a blueprint for a new class of financial co-pilots that integrate quantitative rigor with interactive, cognitively-aware guidance, enhancing both decision quality and user trust.

CCS Concepts

• **Social and professional topics** → **Economic impact**.

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Generative large language models (LLMs), Multi-agent systems, Behavioral finance, Portfolio optimization, Personalized advisory

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1 Introduction

Over the past decade, rising macroeconomic volatility and rapid fintech innovation have fueled unprecedented demand for hyper-personalized wealth management solutions. By 2023, global assets under management (AUM) had reached USD 1.2 trillion [25], yet an estimated 320 million potential investors remained underserved by conventional advisory channels. This gap is especially acute in China, where a burgeoning “new middle class” with rising investable assets is increasingly seeking technology-enabled financial guidance. Empirical evidence from 290 Chinese cities shows that fintech adoption significantly enhances financial access, depth, and inclusion—particularly benefiting households with lower wealth and limited access to traditional advisory services [14].

Traditional wealth management frameworks are under growing pressure to meet the evolving expectations of an increasingly diverse and digitally sophisticated investor base. In markets such as China, the dominance of bank-led fund distribution channels—typically associated with high fees and limited personalization—further constrains access to tailored services, particularly for emerging investors and the expanding middle class. Beyond structural inefficiencies, behavioral challenges such as speculative trading, early fund redemptions, and limited financial literacy remain widespread. Trust in conventional advisory models is further eroded by inadequate transparency and a lack of timely, proactive engagement [13].

Moreover, most incumbent platforms lack the real-time, behavior-aware analytics required to support investors during volatile market conditions, ultimately failing to address both performance-driven goals and the psychological factors that shape retail investment behavior [14].

To address these challenges, recent research has increasingly focused on emerging technologies that enable more adaptive and investor-centric solutions. Advances in large language models (LLMs) and multi-agent frameworks offer promising alternatives to conventional financial advisory systems. LLM-based advisors facilitate personalized, context-aware interactions [21, 27], yet remain costly to deploy and limited in adaptability [7]. Multi-agent systems improve scalability by distributing financial tasks across specialized roles [10, 20]. However, most implementations exhibit three key limitations. First, the lack of behavioral modeling prevents these systems from accounting for cognitive biases—such as overconfidence or loss aversion—that frequently shape retail investor behavior. Second, in the absence of interactive explainability, users receive opaque recommendations that undermine trust and reduce engagement. Third, the static nature of their decision logic impedes real-time responsiveness to market events—an essential capability for maintaining portfolio relevance in volatile environments.

Given these limitations in existing systems, the choice of an underlying investment philosophy is critical. We anchor our framework in **asset allocation**, which financial literature widely recognizes as the cornerstone of long-term portfolio returns, responsible for over 90% of performance variance [4, 5, 12]. In contrast to more speculative and volatile strategies like market timing or stock picking, asset allocation provides a more stable, risk-managed path to wealth growth through diversification and systematic risk management. This philosophy is implemented through two complementary layers: **Strategic Asset Allocation (SAA)**, which sets the foundational, long-term policy portfolio, and **Tactical Asset Allocation (TAA)**, which enables dynamic, short-term adjustments to that policy. The structured yet flexible nature of the SAA-TAA framework makes it an ideal domain for an advisory system that must also address the critical gaps in behavioral modeling and real-time adaptiveness.

We present **Wealth-Voyager**, a behavior-aware, multi-agent financial advisory framework coordinated by a central LLM-based meta-controller. The system integrates structured financial signals, unstructured market narratives, and investor-specific behavioral traits to deliver adaptive, personalized, and interpretable portfolio guidance. Through structured function calls and shared memory, the LLM orchestrates a team of specialized agents responsible for user profiling, news interpretation, portfolio optimization, and advisory dialogue simulation. By embedding Monte Carlo-based risk modeling, event-driven rebalancing, and cognitive bias adaptation, **Wealth-Voyager** moves beyond static allocation tools. It functions as an interactive co-pilot, enabling users to understand, trust, and iteratively refine their financial decisions in dynamic market environments.

Key Contributions:

- **Modular multi-agent collaboration.** **WealthVoyager** orchestrates four specialised agents—**AssistHub** (behavioural profiling),

NewsCrawler (real-time intelligence extraction), **AlphaForge** (scenario-aware portfolio optimization), and **DualAdvisor** (BDI-based advisory simulation)—to emulate the workflow of elite investment-research teams.

- **Behaviour-aware personalization.** Leveraging LLM-based semantic reasoning, **AssistHub** infers latent client preferences and cognitive biases, enabling personalised strategies aligned with individual behavioural profiles that adapt and improve over time.
- **Real-time adaptive decision-making.** By continuously monitoring market dynamics through **NewsCrawler** and performing on-the-fly re-optimization in **AlphaForge**, the system reduces decision latency, supports timely portfolio rebalancing, and proactively mitigates the impact of black-swan events.
- **Bridging the education–decision gap.** Through explainable dialogue and interactive scenario testing, **DualAdvisor** aligns financial education with actionable portfolio choices, empowering users to develop long-term wealth-management competence.
- **Transparency and trust by design.** Through scenario-based analysis, rationale tracing, and verifiable neutral recommendations, the system reveals its decision-making logic, promotes financial literacy, and builds user trust in a transparent and accountable manner.

Collectively, **WealthVoyager** introduces a promising framework that transitions from static asset allocation strategies to an interactive system capable of delivering personalized, adaptive, and transparent wealth management at scale.

2 Related Work

2.1 Multi-Agent Frameworks

Multi-agent (MA) frameworks have gained increasing traction in financial decision-making by distributing portfolio management tasks among specialized agents, each responsible for subtasks such as asset selection, market monitoring, or risk control. Recent systems such as *FinCon* [29] incorporate large language models (LLMs) as high-level coordinators across agent teams, facilitating natural language-based communication and reflective belief revision in volatile environments. *MASA* [20] introduces a role-based structure comprising reward maximizers, risk solvers, and macro observers, with each agent adapting autonomously through deep reinforcement learning. Earlier frameworks, including *MAPS* [18] and *MSPM* [11], emphasized modularity by decoupling strategy generation from asset evaluation, thereby enhancing diversity and scalability. More recent efforts, such as *MASS* [10] and *TradingAgents* [28], simulate collaborative trading desks where LLM-driven agents assume specialized roles—such as sentiment analysis and trade execution—demonstrating strong performance in high-frequency trading environments.

While existing systems demonstrate strong specialization, they often focus on short-term trading or static planning, with limited support for continuous adaptation or investor-facing education. **Wealth-Voyager** addresses these gaps by enabling persistent, behavior-aware coordination that evolves with user preferences and market dynamics. Leveraging real-time news analytics and a Monte Carlo re-sampling loop, it can anticipate rare but disruptive events such as market shocks. Meanwhile, the **DualAdvisor**

module fosters financial literacy through interactive dialogues that clarify trade-offs, surface biases, and enable scenario-based learning—shifting the system from execution to long-term investor empowerment.

2.2 Large Language Models for Investment Decision Support

Large language models (LLMs) have become increasingly influential in financial decision support by enabling contextual understanding, generation, and interaction across a range of investment-related tasks. *FinLLM* [21] applies instruction tuning to provide personalized advice to retail investors, while *BloombergGPT* [27] demonstrates the advantages of large-scale, domain-specific pretraining for tasks such as entity recognition and financial question answering. *LLM-Advisor* [16] integrates real-time news summarization and event interpretation into an interactive advisory workflow, and *LLM-Survey* [7] offers a comprehensive overview of how LLMs support the asset management lifecycle—from information extraction to portfolio construction. Despite this progress, many existing systems under-address critical challenges related to probabilistic uncertainty and behavioral biases in investor decision-making.

Despite recent advances, many LLM-based financial systems still struggle with behavioral uncertainty, limited transparency, and high deployment costs. While *LLM-Advisor* incorporates real-time data and *BloombergGPT* scales well for domain coverage, such systems often operate as single-pass generators without introspection or adaptive feedback. **Wealth-Voyager** introduces a cost-efficient, dialogue-centric interface grounded in behavioral finance, where a belief–desire–intention (BDI) framework supports explainable, bias-aware recommendations. By simulating advisor–client interactions, the system uncovers cognitive biases and promotes reflective decisions through transparent strategy rationales.

2.3 Traditional and Behaviour-Aware Approaches to Portfolio Optimization

Traditional portfolio optimization techniques, such as mean–variance analysis and Conditional Value at Risk (CVaR), remain foundational to financial planning [9]. Monte Carlo simulation (MCS) has further enabled probabilistic outcome estimation, although early implementations are generally offline and single-pass. *MCS-Interactive* [3] introduces conversational simulation, while our system (see Appendix A.3) advances this line by supporting custom asset universes and user-defined risk thresholds, coupled with an LLM-driven re-sampling loop for adaptive portfolio generation.

Beyond numerical optimization, behavior-aware systems incorporate cognitive biases—such as loss aversion and overconfidence—that influence real-world investor decisions. *Cognitive-Aware Advisors* [6, 8] demonstrate that embedding behavioral profiling improves trust and engagement. Our approach adopts a nine-dimensional bias vector (see Appendix A.2) to guide LLM-based advisory dialogue, aligning with findings from *Trust-Aware GAI Advisors* [26] on personalized, trust-sensitive interaction.

3 Framework

3.1 Overall Architecture

To provide personalized and real-time financial advisory services for individual users, we propose an end-to-end framework named **Wealth-Voyager**. As illustrated in Fig. 1, the system integrates multiple collaborating agents that dynamically interact with core tools, including an intelligent customer assistant, a news crawler, and portfolio optimization models. In addition, a role-playing dialogue module that employs a dual-agent, role-playing simulation to deliver adaptive guidance that is responsive to both real-time market events and the user’s unique behavioral profile.

3.2 Core Modules

AssistHub is an intelligent customer service module, implemented as an agentic workflow on the Dify [17] platform and powered by ChatGPT-4 [1]. Through a guided conversational process of context-aware, adaptive testing and interactive calibration, it dynamically handles general inquiries and assists users in articulating their wealth management goals (e.g., retirement savings, educational funds), detailing their current financial status, and completing a personalized risk assessment. AssistHub replaces the “static questionnaire” with a guided multi-round dialogue mechanism to comprehensive user profile. This profile, which encapsulates the user’s financial status, long-term goals, and risk tolerance, serves as a foundational input required by the other specialized modules within the **Wealth-Voyager** framework to deliver personalized analysis and advice.

NewsCrawler operates as a dedicated Model Context Protocol (MCP) [22] server responsible for the real-time aggregation and processing of market information from a wide array of online sources. This module serves two critical functions within the **Wealth-Voyager** framework. Firstly, it periodically extracts expected return forecasts for major asset classes from research reports published by professional financial institutions. This data provides the **AlphaForge** module with realistic and timely inputs, significantly enhancing the practical applicability and accuracy of the portfolio optimization algorithms. Secondly, NewsCrawler curates and retrieves daily market news tailored to each user’s specific profile, including their stated investment interests, risk tolerance, and existing portfolio composition. This personalized news feed serves as the foundational context for the role-playing dialogues within the **DualAdvisor** module.

AlphaForge is the quantitative core of **Wealth-Voyager** which is based on morden portfolio theory. Drawing on market expectations (returns, volatilities, and cross-asset correlations) continuously harvested by **NewsCrawler** and the investor-specific constraints produced by **AssistHub**, it constructs a long-term strategic asset-allocation that balances growth, risk, and liquidity. Internally, AlphaForge employs a constrained mean–variance engine augmented with liquidity caps, draw-down limits, and leverage controls. The solver outputs target weights alongside forward-looking metrics—projected return, volatility, Value-at-Risk, and liquidity coverage—that feed downstream advisory logic. The full mathematical specification and optimization routine are provided in Appendix A.3

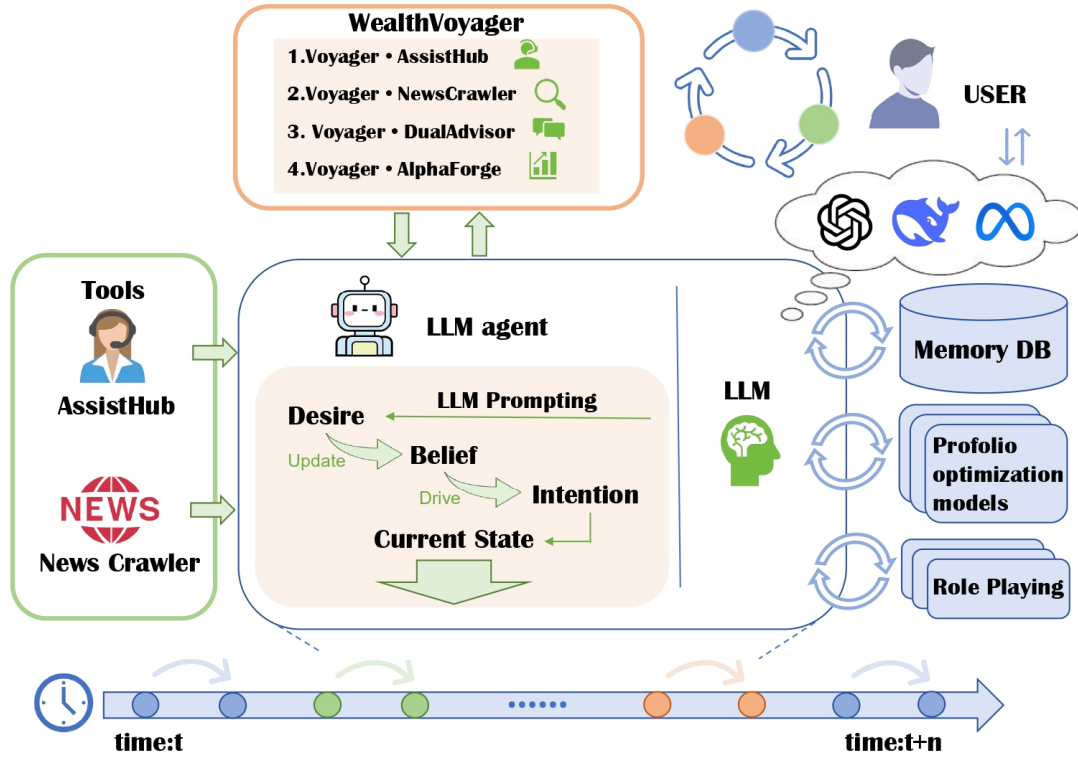


Figure 1: The architecture of the proposed Wealth-Voyager framework.

DualAdvisor module utilizes a novel role-playing framework powered by two distinct large language model (LLM) agents, which simulate consultations between a professional investment advisor and the user. The primary purpose of this module is to generate real-time, tactical investment advice. By dynamically modeling advisor-client interactions in response to market events, DualAdvisor provides actionable recommendations that help users make timely portfolio adjustments to navigate market volatility effectively.

3.3 Role-Playing-Based Advisory Simulation

Traditional financial advisory systems typically rely on static user profiles, which fail to capture the dynamic interplay between evolving market conditions and an investor’s real-time cognitive and emotional responses [24]. To address this limitation, we propose **DualAdvisor**, a role-playing module within **Wealth-Voyager**, designed to simulate multi-turn, psychologically informed dialogue between a professional advisor and a digitally modeled client. The objective is to move beyond static recommendations by generating advisory responses that are both responsive to market signals and adaptive to user-specific behavioral biases. This is achieved through an agent-based interaction between two large language model (LLM) agents, grounded in a behaviorally rich user profile.

3.3.1 Agent Roles and Dialogue Objectives. To effectively simulate complex financial decision-making, we decompose the advisory interaction into two specialized agent roles, inspired by recent developments in social-agent simulation [19].

- **Advisor Agent:** Acts as the rational anchor. It interprets market developments with professional neutrality and proactively detects behavioral biases reflected in the user’s responses. For example, it may caution against overreaction to news due to high policy sensitivity or discourage impulsive “gain-chasing” behavior. The Advisor continuously steers the dialogue toward strategies consistent with the user’s stated long-term financial goals.
- **User Agent:** Serves as a high-fidelity “digital twin” of the user, designed to inject psychologically plausible, and often biased, human behavior into the simulation. Its purpose is not to be perfectly rational but to realistically model how a specific individual would react to market changes based on their unique financial circumstances, goals, and behavioral tendencies.

Each agent is instantiated using *role-specific system prompts*. The User Agent prompt dynamically incorporates the complete user profile, including financial goals, risk tolerance, and quantified behavioral attributes, thereby ensuring psychologically coherent and context-aware responses.

This dual-agent design fulfills two complementary objectives. Firstly, it generates personalized, bias-aware investment guidance by anticipatorily modeling the user’s likely emotional reactions. Secondly, the recorded conversation acts as a “behavioral mirror” for investor education, enabling users to observe how professional reasoning can mitigate their own cognitive distortions, thereby enhancing their financial self-awareness over time.

3.3.2 BDI-Grounded Behavioral Profile. To enable realistic simulation, we construct a comprehensive user profile that serves as a behavioral fingerprint. Drawing from behavioral economics, this profile extends beyond conventional data by quantifying a range of user-specific biases. These are broadly categorized into **emotional indicators** (e.g., loss aversion, overconfidence), which model the user’s core feelings about risk and security, capturing the affective responses under market stress that often drive intuitive decisions, and **cognitive indicators** (e.g., herding tendency, policy sensitivity), which simulate how the user processes market information and forms judgments, reflecting their systematic patterns of reasoning.

To operationalize this behavioral profile within the User Agent, we model its internal deliberation using the Belief–Desire–Intention (BDI) framework, a well-established paradigm for rational agency that mirrors human practical reasoning. This structure sequentially models the agent’s:

- **Beliefs (B):** Its subjective understanding of the current market.
- **Desires (D):** Long-term investment goals and emotional inclinations derived from the user profile.
- **Intentions (I):** Actionable steps proposed for portfolio adjustment.

We operationalize BDI reasoning via a *step-wise prompting* mechanism: the LLM is sequentially queried for its beliefs, desires, and intentions in response to new market information. This chain-of-thought design improves both the logical consistency and interpretability of the simulated decision process.

3.3.3 Dialogue Flow and Consensus Mechanism. The agent interaction unfolds as a structured, multi-turn consultation designed to transform a user’s potential biased reactions into a reasoned action plan. The dialogue progresses through four distinct phases:

- **Rational Initialization:** The Advisor Agent synthesizes daily market news through the lens of the user’s profile to formulate a scenario-specific interpretation. This interpretation serves as the initial prompt, triggering the User Agent’s BDI reasoning process.
- **Behavioral Response:** The User Agent then generates its initial BDI response, surfacing the user’s likely first reaction, complete with any possible biases.
- **Cognitive Debiasing:** The Advisor Agent critiques this response, identifies irrational patterns based on the user’s profile, and recommends alternative, goal-aligned plans.
- **Iterative Consensus:** The agents engage in an iterative dialogue. The User Agent either revises its intentions based on the Advisor’s feedback or defends its position to prompt further clarification. The dialogue terminates when a consensus is reached, governed by a pragmatic mechanism: either the Advisor generates a specific agreement phrase (e.g., “Agree, no modification needed”) or a predefined maximum of ten rounds is reached.

The entire interaction culminates in a final, user-facing deliverable: the *Daily Investment Briefing*. This structured report summarizes the essence of the dialogue, providing the user with a clear and actionable summary that includes a market overview, the Advisor Agent’s tailored interpretation, concrete portfolio adjustment

suggestions derived from the consensus, and key areas of focus for the near future.

4 User Study

To evaluate the practical effectiveness and decision-support capabilities of the **Wealth-Voyager** system, we conducted a month-long pilot study from April 2 to May 2, 2025. The study involved a mid-career participant focused on retirement planning and aimed to simulate realistic investment scenarios under live market conditions. During this period the system was fully deployed in a sandbox environment, enabling end-to-end validation of its core modules—including behavioural profiling, adaptive rebalancing, and agent-driven market response.

Importantly, the experiment coincided with a period of elevated geopolitical and financial uncertainty, marked by a sharp escalation in global trade tensions. This provided a natural stress-test for evaluating how **Wealth-Voyager** responds to volatile market events.

The study focus on the quantitative performance (e.g. returns and volatility) of both Strategic Asset Allocation (SAA), which based on long-term investment goals and Tactical Asset Allocation (TAA), which based on SAA and dynamically adjusted according to short-term market signals (such as policy changes, macroeconomic data), and qualitative user experience offering an integrated view of the robustness, adaptability, and usability of our system.

4.1 Participant Profile

The case involves a 45-year-old professional planning to retire within eight years, with the objective of growing investable assets to support a comfortable post-retirement lifestyle. During onboarding, the user reported substantial wealth and moderate liquidity requirements but did not specify key parameters such as risk tolerance or return objectives. The initial portfolio allocation (Table 1a) includes a 20% concentration in real estate investment trusts (REITs)—an unusually high exposure for a balanced portfolio—introducing liquidity constraints and limiting diversification.

Onboarding interactions revealed three primary concerns: (i) mitigating large drawdowns near retirement, (ii) maintaining access to capital within six months for potential obligations, and (iii) receiving clear, rationale-driven investment guidance. These concerns informed the system’s evaluation criteria, emphasizing drawdown control, risk-adjusted performance, and interpretability.

In contrast to conventional systems that provide static recommendations or opaque optimizations, **Wealth-Voyager** inferred latent preferences and applied a structured, investment-bank-style workflow to revise the portfolio. The resulting allocation (Table 1b) improved diversification and more closely aligned with the user’s implicit risk profile—demonstrating **Wealth-Voyager**’s capacity to translate vague financial goals into actionable, personalized investment strategies.

4.2 System Operation

To ensure timely adaptation to market shifts, the system continuously scraped macroeconomic news and monitored real-time asset prices. The **AlphaForge** firstly implements Strategic Asset Allocation (SAA) by optimizing long-term weights under constraints such

Table 1: Comparison between user-declared baseline profile and AI-optimised allocation.

Attribute	User-declared (Apr 2, 2025)	AI-optimised (DUALADVISOR)
Initial capital	RMB 3,000,000	RMB 3,000,000
Target amount	RMB 4,500,000+	RMB 4,500,000+
Risk tolerance	N/A	Medium (max loss 20%)
Leverage allowed	No	No
Liquidity need	Medium	Medium
Strategic asset mix	A-shares 30%, Bonds 35%, Gold 15%, REITs 20%	A-shares 35.3%, Bonds 41.1%, Gold 17.6%, REITs 6.0%
Expected return	N/A	6.53%
Expected volatility	N/A	9.42%

Notes. “A-shares” are RMB-denominated equities on mainland Chinese exchanges; “Bonds” are fixed-income securities; “Gold” includes physical bullion or ETF exposure; “REITs” are real-estate investment trusts. N/A = not provided at onboarding.

as maximum drawdown thresholds. Additionally, the **NewsCrawler** module generated personalised briefings and contextual interpretations tailored to the participant’s portfolio exposures. Upon detection of disruptive events—such as unexpected policy shifts or volatility spikes—the **DualAdvisor** module assessed potential portfolio impacts and issued preemptive rebalancing proposals to hedge downside risk while preserving upside potential and also employs Tactical Asset Allocation (TAA) to adjust positions in response to real-time market events with the algorithm in the **AlphaForge**.

On April 2, 2025, a sudden tariff shock (a minor “black-swan” event) was detected. Tactical adjustments were executed prior to significant market drawdowns. These included a reduction in equity holdings, increased exposure to gold and government bonds, and a moderate uptick in REIT allocation. This repositioning enabled the participant to limit losses and capitalise on short-term mispricings.

4.3 Performance Across Portfolio Strategies

Figure 2 compares segment-level returns of passive and tactical portfolios from April 2 to May 9, 2025. Following a tariff-induced shock (Apr 02–Apr 09), the tactical approach reduced losses (−2.56% vs. −2.79%). In the rebound phase (Apr 09–Apr 23), it captured greater upside (3.24% vs. 2.16%). Across subsequent intervals, tactical returns remained consistently higher, notably during Apr 30–May 07 (1.15% vs. 0.68%). Even during mild pullbacks (May 07–May 09), the tactical strategy limited downside more effectively.

These results underscore the advantage of layering Tactical Asset Allocation (TAA) atop a Strategic Asset Allocation (SAA) framework, enhancing responsiveness without sacrificing stability.

Table 2 compares the performance of three portfolio strategies: the user-declared baseline (Original), a behaviorally *Anchored* portfolio, and a dynamically rebalanced *Tactical* strategy. Anchoring alone yields substantial gains, increasing the annual return from 3.72% to 6.53% while nearly halving annualised volatility (from 18.08% to 9.42%).

Building on this foundation, the Tactical strategy delivers further improvements. Over the evaluation period, cumulative return rises from 0.24% to 1.86%, with annual volatility reduced to 12.10%, outperforming both baseline and Anchored strategies on a risk-adjusted basis.

These results validate the effectiveness of **WealthVoyager’s** layered decision architecture: the integration of strategic behavioral

anchoring with real-time tactical rebalancing achieves superior returns without increasing risk exposure.

4.4 Comparative Analysis with Existing Market Solutions

Recent industry analyses highlight the growing adoption of AI-driven wealth-management systems in China, notably: Ant Group’s *Ant Fortune* [2], JD Digits’ *JD Xiaobei* [15], and Ping An’s *Intel-ligent Wealth Butler* [23]. These platforms provide foundational investment guidance through rule-based allocation and basic user profiling.

In contrast, **Wealth-Voyager** was co-developed with a group of novice and intermediate investors who used it alongside existing market solutions over a six-week period. Feedback from this real-world deployment informed the design of key differentiators: real-time adaptive recommendations, personalized behavioral modeling, and conversational explanation mechanisms.

Wealth-Voyager not only supports standard investment advisory functions but also introduces novel dimensions that enhance the user experience. As shown in Table 3, our system outperforms existing offerings in areas such as dynamic responsiveness, personalization depth, and educational engagement — features that users highlighted as critical for building long-term financial literacy and confidence.

The comparison reveals that while current mainstream platforms fulfill baseline advisory requirements, they remain limited in their responsiveness and personalization depth. Users reported that **Wealth-Voyager’s** ability to explain investment logic in plain language and adapt to their evolving preferences significantly improved their engagement and understanding of portfolio management.

Wealth-Voyager advances the paradigm by enabling adaptive intraday asset rebalancing, integrating behavioral finance signals, and providing interpretable recommendations via dialogue-based interfaces. These enhancements collectively position it as a more transparent and responsive co-advisor across diverse investor segments.



Figure 2: Segment-level returns for passive and tactical portfolios (April 2–May 9, 2025).

Table 2: Performance comparison across user-declared, anchored, and tactical portfolio strategies.

(a) Original vs. Anchored			(b) Anchored vs. Tactical		
Portfolio	Annual Return (%)	Annual Volatility (%)	Portfolio	Cumulative Return (%)	Annual Volatility (%)
Original	3.72	18.08	Anchored	0.24	13.70
Anchored	6.53	9.42	Tactical	1.86	12.10

Notes. *Original*: user-declared allocation; *Anchored*: allocation anchored by behavioral signals via **DualAdvisor**; *Tactical*: adaptively rebalanced based on live market signals.

Annual return = projected yearly gain; cumulative return = total return over the observed period; annual volatility = yearly return dispersion.

Table 3: Capability comparison between Wealth-Voyager and representative market offerings. ✓ = supported; ✗ = absent.

Feature	Ant Fortune	JD Xiaobei	Ping An Butler	Wealth-Voyager
Diverse asset-class support	✓	✓	✓	✓
Interactive interface	✓	✓	✓	✓
Real-time dynamic adjustment	✗	✗	✗	✓
Personalized customization	✗	✗	✗	✓
Financial education integration	✗	✗	✗	✓

5 Conclusion

In this paper, we introduced **Wealth-Voyager**, a novel multi-agent framework that unifies behavioural profiling, real-time market intelligence, and quantitative optimization into a cohesive pipeline. Through the coordination of four specialized modules—**AssistHub**, **NewsCrawler**, **AlphaForge**, and **DualAdvisor**—our system delivers interactive decision support that moves beyond passive recommendations. This was evidenced by our proof-of-concept deployment, where the system’s adaptive advice outperformed a passive baseline by +1.62 pp during a period of elevated macro uncertainty, and its strategic allocations improved the user’s risk-return trade-off (+2.81 pp annualized return, −8.66 pp volatility). Qualitative feedback also indicated that the dual-agent simulation enhances

user trust and self-awareness by exposing cognitive biases in an explainable dialogue.

Our validation, while promising, has several limitations: the prototype was evaluated on a single user and a limited asset universe, and its performance is contingent on the underlying LLM. To address these gaps and strengthen the robustness of our findings, our future work will prioritize a multi-faceted evaluation. We plan to conduct larger, longitudinal studies with a diverse cohort of users to statistically validate the system’s adaptability across different behavioral profiles. Furthermore, we will expand the asset universe, test the framework under varied market regimes to assess its resilience, and perform systematic benchmarking against state-of-the-art systems to highlight the advantages of our behavior-aware

approach. Beyond these evaluations, we also aim to enhance the core methodology by exploring reinforcement learning to refine the negotiation dynamics within DualAdvisor and investigating domain-specific models to improve performance and reduce inference costs.

By bridging the gap between behavioural finance, quantitative analysis, and generative AI, **Wealth-Voyager** charts a viable path toward the next generation of financial co-pilots. We believe this work represents a significant step toward more effective, transparent, and trustworthy human-AI collaboration in the complex domain of wealth management.

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A Appendix

A.1 Monte-Carlo-Driven Meta-Agent

Algorithm 1 Investment Advisor Loop (simplified)

Require: User config c_0 , success threshold τ , max rounds R

```

1:  $c \leftarrow c_0$ ;  $best_c \leftarrow c_0$ ;  $best_p \leftarrow 0$ 
2: for  $r \leftarrow 1$  to  $R$  do
3:    $p \leftarrow \text{MONTECARLOSIM}(c)$ 
4:   if  $p > best_p$  then
5:      $best_c \leftarrow c$ ;  $best_p \leftarrow p$ 
6:   end if
7:   if  $p \geq \tau$  then
8:     break
9:   end if
10:   $c \leftarrow \text{LLM\_ADJUST}(c, \text{allowed\_params})$ 
11: end for
12: return ( $best_c$ ,  $best_p$ )
```

Function Descriptions for Algorithm 1 (Investment Advisor Loop):

- **MONTECARLOSIM(c)**: runs a Monte Carlo rollout under configuration c and returns the success probability estimate.
- **LLM_ADJUST(c, allowed_params)**: interacts with an LLM-based dialogue agent to propose adjustments to the current user configuration c , subject to a whitelist of allowed parameters.

Algorithm 2 Construction of Behaviour Vector

Require: Conversation ID id ; default mapping $\mathcal{M}$ (Table 4)

```

1:  $H \leftarrow \text{LOADDIALOGUEHISTORY}(id)$ 
2: if CONTAINSSTRUCTUREDBLOCK( $H$ ) then
3:    $E \leftarrow \text{PARSESTRUCTUREDBLOCK}(H)$ 
4: else
5:    $E \leftarrow \text{LLM\_PROFILEEXTRACT}(H)$ 
6: end if
7: if  $E$  incomplete then
8:    $E \leftarrow \text{FILLDEFAULTS}(E)$ 
9: end if
10:  $p \leftarrow E.\text{investment\_purpose} \triangleright \text{retirement} / \text{education} / \text{house} / \text{growth}$ 
11:  $v \leftarrow \mathcal{M}[p]$ 
12:  $v[4] \leftarrow E.\text{real\_time\_emotion}$ 
13: return DICT(keys, v)

```

A.2 Large-Model-Assisted Behaviour Profiling

Function Descriptions for Algorithm 2 (Behaviour Vector Construction):

- **LOADDIALOGUEHISTORY(id):** retrieves the entire conversation history for a given conversation ID.
- **CONTAINSSTRUCTUREDBLOCK(H):** checks if a well-formed, machine-readable block is present in the dialogue.
- **PARSESTRUCTUREDBLOCK(H):** parses a structured block to directly extract user profile attributes.
- **LLM_PROFILEEXTRACT(H):** uses a language model to infer user attributes from free-form dialogue if no structured block is found.
- **FILLDEFAULTS(E):** fills missing entries in the extracted profile vector E using predefined default values.

Behavioral Metric Descriptions:

- **Loss Aversion:** This is an emotional bias where investors tend to feel the pain of losses more than the pleasure of gains.
- **News & Policy Sensitivity:** This metric gauges how much an investor's decisions are irrationally swayed by immediate media headlines and policy changes.
- **Investment Experience:** An investor's behavior is unconsciously influenced by past experiences, which can provide more accurate insight into their risk profile.
- **Real-Time Emotion:** This refers to reasoning influenced by feelings, which can overpower logical thinking during times of stress and affect investment decisions.
- **Herding Tendency:** This is a behavior where investors do not have their own ideas and instead follow the lead of their friends and colleagues in making investment decisions.
- **Regret Aversion:** This is an emotional bias where investors avoid taking decisive actions because they fear that their chosen course will prove unwise in hindsight.
- **Overconfidence:** This is an emotional bias best described as unwarranted faith in one's own thoughts and abilities.
- **Illusion of Control:** This is a cognitive bias where people believe they can control or influence investment outcomes when they actually cannot.

- **Decision Delay:** This is a behavior where investors are slow to make decisions due to discomfort with uncertainty or putting off decisions until they receive professional advice.

Table 4: Default Personality Mapping $\mathcal{M}$

Purpose	Vector slice (loss_aversion, ..., decision_delay)
retirement	[0.8, 0.5, 0.8, 0.5, 0.4, 0.8, 0.3, 0.2, 0.6]
child_education	[0.7, 0.6, 0.5, 0.5, 0.5, 0.7, 0.4, 0.4, 0.5]
house_purchase	[0.6, 0.7, 0.4, 0.5, 0.6, 0.6, 0.3, 0.3, 0.7]
wealth_growth	[0.3, 0.8, 0.6, 0.5, 0.7, 0.3, 0.8, 0.7, 0.3]

A.3 Constraint-Aware Portfolio Optimiser**Algorithm 3** Multi-Stage Optimization Procedure

Require: Mean returns μ , covariance Σ , user config κ

```

1: Infer target return  $r^*$ , vol limit  $\sigma_{\max}$ , CVaR limit
2: Derive liquidity tiers, leverage flag and restricted set  $\mathcal{R}$ 
3: Build box bounds  $b_i$  for all assets
4:  $w^{(0)} \leftarrow \text{ERC\_INIT}(\Sigma)$ 
5:  $w_{\text{MS}} \leftarrow$  closed-form max-Sharpe solution
6: if  $w_{\text{MS}}$  feasible and meets all risk caps then
7:   return  $w_{\text{MS}}$ 
8: end if

```

$\triangleright$ — Stage 1: smooth local search —

```

9:  $\mathcal{L}(w) \leftarrow -\text{Sharpe}(w) + \lambda \text{IlliqPenalty}(w)$ 
10:  $w^* \leftarrow \text{SLSQP}(\mathcal{L}, w^{(0)}, b_i, \text{constraints})$ 
11: if SUCCESS then
12:   return POSTPROCESS( $w^*$ )
13: end if

```

$\triangleright$ — Stage 2: global repair —

```

14:  $w^\diamond \leftarrow \text{DIFFERENTIALEVOLUTION}(\mathcal{L}, b_i, \text{constraints})$ 
15: if SUCCESS then
16:   return POSTPROCESS( $w^\diamond$ )
17: else
18:   return FAILREPORT()
19: end if

```

Function Descriptions for Algorithm 3 (Multi-Stage Optimization):

- **ERC_INIT(Σ):** returns an equal risk contribution portfolio initialization given the covariance matrix Σ .
- **SLSQP($\mathcal{L}, w^{(0)}, b_i, \text{constraints}$):** applies Sequential Least Squares Quadratic Programming to locally optimize the objective $\mathcal{L}$ starting from $w^{(0)}$.
- **DIFFERENTIALEVOLUTION($\mathcal{L}, b_i, \text{constraints}$):** applies a global evolutionary search to optimize $\mathcal{L}$ in case the local method fails.
- **POSTPROCESS(w):** normalizes and sanitizes a candidate weight vector before returning it.
- **FAILREPORT():** generates a diagnostic message if no feasible solution is found.

A.4 Illustrative Case Study: A Tactical Response Narrative

To illustrate the practical utility of the framework, this appendix offers a narrative walkthrough of a five-stage tactical response to a simulated market crisis. The box below summarizes the key events, corresponding analyses, and portfolio adjustments made at each stage.

Investment Strategy Review and Outlook

Phase 1: Crisis Response (Apr 2: Tariff Shock)

News: On April 2, the U.S. President announced a "Liberation Day" tariff policy, triggering a global stock market sell-off.

Analysis: The market experienced a sudden downturn. For a client with a "retirement" objective, the primary task is to strictly control portfolio drawdown and protect hard-earned capital. We must adopt decisive defensive measures to prevent panic from affecting the long-term plan.

Portfolio Adjustment:

A-Shares	Bonds	Gold	REITs
27.9% ↓	26.6% ↓	38.0% ↑	7.5% ↑

Phase 2: Market Observation & Final Hedging (Apr 8: Pre-Rebound Signals)

News: On April 8, "pre-rebound signals" emerged as the stock market saw a minor recovery and the CBOE Volatility Index (VIX) stabilized.

Analysis: The market decline has paused, but the true trend remains uncertain. Such a fragile rebound is not worth the risk. Maintaining caution and completing the final defensive adjustments are crucial to ensure we are in the safest position before the situation fully clears, allowing us to seize the initiative later.

Portfolio Adjustment:

A-Shares	Bonds	Gold	REITs
18.9% ↓	34.3% ↑	41.3% ↑	5.5% ↓

Phase 3: Decisive Re-entry (Apr 9: Tariff Suspension)

News: On April 9, the suspension of some tariffs was announced, leading to a rapid recovery in global risk assets.

Analysis: This definitive positive news has turned the market's direction. While our previous defensive posture was successful, we must now respond swiftly. To avoid missing the core of this rally, we need to act immediately and reinvest capital into the A-share market, a key offensive move in our "phased re-entry" plan.

Portfolio Adjustment:

A-Shares	Bonds	Gold	REITs
36.0% ↑	24.9% ↓	31.4% ↓	7.3% ↑

Phase 4: Trend Confirmation & Steadiness (Apr 11: Consumer Confidence Drop)

News: On April 11, the University of Michigan Consumer Sentiment Index fell to a near-record low, creating mixed market signals.

Analysis: Negative macroeconomic news serves as a reminder that the market's foundation is not yet solid. This confirms the wisdom of our earlier decision not to go all-in at once. During such periods of mixed signals, the best strategy is to remain patient, maintain the current allocation, and observe further developments.

Portfolio Adjustment:

A-Shares	Bonds	Gold	REITs
36.0% (Hold)	24.9% (Hold)	31.4% (Hold)	7.3% (Hold)

Phase 5: Crisis Aversion & Normalization (May 2: U.S. Stocks Recover)

News: On May 2, U.S. stocks erased all "Liberation Day" losses.

Analysis: The market turmoil has subsided, and your portfolio has successfully passed this stress test. Our "defense first, offense later" tactic proved effective. The market's focus now returns to fundamentals, and so our focus must shift from short-term responses back to your eight-year retirement objective.

Portfolio Adjustment:

A-Shares	Bonds	Gold	REITs
36.0% (Hold)	24.9% (Hold)	31.4% (Hold)	7.3% (Hold)

B Research Methods

B.1 Part One

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B.2 Part Two

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C Online Resources

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